DAIGNOSIGN HARDWARE FAILURES

1 List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

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Hardware Issue	Typical Symptom
1. RAM failure	PC does not boot, blank screen, or repeated beep sounds.
2. Hard Drive/SSD failure	PC becomes very slow, files cannot open, or OS does not boot.
3. Power Supply failure	PC does not turn on or suddenly shuts down.
4. Overheating CPU	PC becomes slow, freezes, or shuts down automatically.
5. Graphics Card failure	No display, screen artifacts, or display crashes.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

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- **Check the charger and cable** – Look for cuts, bends, loose connections, or damage.
- **Check the power socket** – Plug another device into the socket to confirm it is working.
- **Check the charging port** – Make sure the laptop port is clean and the charger fits properly.
- **Check the charger output with a multimeter** – Measure the adapter voltage and compare it with the voltage printed on the charger.
- **Try another compatible charger** – This helps identify whether the original charger is faulty.
- **Remove and reconnect the battery (if removable)** – Check whether the battery connection is loose or damaged.
- **Inspect the battery** – Check for swelling, leakage, or physical damage. **Do not use a swollen battery.**

3.Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

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Test	Measured Voltage	Expected Range	Result
USB phone charger	**5.02 V**	**4.75 V – 5.25 V**	**Within expected range – PASS**

Procedure: Set the multimeter to **DC voltage (V⎓)**, place the black probe on the USB ground pin and the red probe on the **+5V/VBUS** pin. A reading around **5 V** means the charger output is normal.

**4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could thiindicate?

<em>Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.**

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ATX Power Supply 12V Rail Test

Steps:

1. Shut down the PC and **unplug it from the wall**.
2. Disconnect the ATX power connector from the motherboard.
3. Set the multimeter to **DC Voltage (V⎓)**.
4. Connect the **black probe to a black wire (GND)**.
5. Touch the **red probe to the yellow wire (+12V)**.
6. Record the voltage shown on the multimeter

Example result:

Test	Measured Voltage	Expected	Result
ATX +12V rail	**12.08 V**	Around 12 V	**Normal**